

Venkata Dhanush Venna

562-613-8219 | dhanushevenna2@gmail.com | [linkedin.com/in/dhanush-venna](https://www.linkedin.com/in/dhanush-venna) | github.com/dhanush-venna

EDUCATION

University of California, Irvine
Bachelor of Science in Computer Science

December 2026

SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, Flask, REST APIs, Figma, Tailwind CSS, PyTorch, scikit-learn, NumPy, Pandas, OpenCV
Tools & Platforms: Git, GitHub, VS Code, Jupyter, Linux, Cursor
Core Concepts: Data Structures & Algorithms, Full-Stack Development, Software Design, Data Preprocessing & Analysis, System Design

PROJECTS

UCI ARC Court Reservation Prototype

Spring 2026

React, TypeScript, Figma, Tailwind CSS

- Served as project manager for a 4-person team, assigning tasks based on individual strengths, reviewing deliverables, tracking milestones, and preventing scope creep to ensure timely project completion.
- Defined and prioritized product requirements, expanding the reservation system to support 8 different sports with features including real-time availability, reservations, check-in, open-court joining, and cancellations.
- Led usability improvements that reduced reservation completion time by 53% (from 85 to 40 seconds) through iterative workflow refinement and user feedback.
- Coordinated the project's quality assurance to ensure the final prototype met functional requirements and aligned with user needs.

SARS-CoV-2 Sequence Alignment Platform

Spring 2026

Python, TypeScript, Flask, React, Biopython, NumPy

- Built a bioinformatics platform to compare SARS-CoV-2 spike protein sequences across Wuhan, Delta, and Omicron variants using global and local sequence alignment algorithms.
- Built an accession-specific preprocessing pipeline that extracted and validated spike coding sequences of 3,822, 3,816, and 3,813 nucleotides, reducing exact dynamic-programming computation from 886 million to 14.6 million matrix cells per alignment.
- Executed 2,016 alignment experiments, including a 36-run canonical benchmark and evaluation across 144 scoring profiles, analyzing alignment scores, runtime, mismatch rates, and gap percentages under multiple scoring configurations.
- Benchmarked global and local alignment algorithms, showing Smith–Waterman averaged only 0.10 seconds slower than Needleman–Wunsch while producing identical summary metrics across all tested spike-sequence configurations.

Malaria Parasite Detection from Blood Smears

Fall 2025

Python, PyTorch, Torchvision, scikit-learn, OpenCV

- Developed an image classification model using MobileNetV2 and EfficientNetB0 to detect malaria-infected blood cells from a Kaggle dataset of 27,000+ microscopy images.
- Fine-tuned pretrained CNN architectures with image preprocessing, normalization, augmentation, and binary cross-entropy optimization to improve classification performance and model generalization.
- Achieved 92.6% validation accuracy, 91.9% precision, 93.8% recall, 92.9% F1-score, and a 0.977 ROC-AUC, demonstrating reliable detection of infected blood cells.
- Analyzed training loss and validation metrics across multiple epochs, reducing training loss from 0.3395 to 0.2636 while increasing training accuracy from 85.5% to 89.5%.

EXPERIENCE

PC Building & Reselling

2023 – Present

Independent Experience

- Diagnosed and resolved hardware, software, driver, and operating system issues across custom desktop PCs.
- Built and configured reliable systems by installing hardware components, setting up software environments, and validating system performance.
- Developed strong technical communication skills by translating complex troubleshooting into clear customer guidance.

CERTIFICATIONS

Google AI Essentials, OpenAI Applied AI Foundations, Anthropic AI Fluency: Framework and Foundations, IBM Project Management Fundamentals